Manpreet Singh

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Education

Wilfrid Laurier University **Sep 2025 – Apr 2027**
Master of Applied Computing (Co-op) GPA: 11.34 / 12

Chitkara University **Jul 2021 – May 2025**
Bachelor of Engineering (Computer Science) GPA: 9.23 / 10

Work Experience

Home Hardware Stores Limited **May 2026 – Present**
eCommerce Full Stack Developer Co-op *St. Jacobs, ON*

- Building frontend features for the **Home Hardware Pro Dashboard** using **Vue.js** and **Svelte**, delivering reusable UI components and user stories across **Agile sprints** for **Pro-enrolled stores**.
- Engineered distributed ETL pipelines using **Microsoft Fabric** and **PySpark** to ingest **250K+ daily POS transactions** across **1,100+ nationwide stores**, systematically extracting offline pricing, quantity, and loyalty (Scene/Moneris) metrics.
- Engineered a cross-cloud ingestion pipeline to compute offline mode prices, storing aggregations in a **Fabric Delta Table** and deployed a serverless **GCP Cloud Function** to perform incremental, timestamp-based syncing into **Cloud SQL**.

CoreAI Technologies **May 2025 – Oct 2025**
Full-Stack Software Developer *Remote*

- Developed a high-performance health analytics dashboard using **Next.js**, **React**, and **TanStack Query**, optimizing API data fetching via custom caching layers to seamlessly visualize complex biomarker trends.
- Designed a comprehensive biomarker categorization system structuring **400+ unique metrics** across cardiovascular, blood, and performance profiles, streamlining backend data retrieval and simplifying frontend user navigation.

SalesCode.ai **Jun 2024 – Dec 2024**
Software Engineer Intern *Gurugram, India*

- Architected a geo-targeted banner delivery system using **Java (Spring Boot)** REST APIs, deployed on an auto-scaling **AWS EC2 and S3** infrastructure to dynamically serve localized content to **100K+ sales representatives**, driving a **15% increase in click-through rates**.
- Spearheaded performance engineering for the **Flutter** mobile client, refactoring widget rendering pipelines and implementing lazy loading patterns to reduce the application memory footprint by **over 30% (from 89MB to 62MB)**.

Projects

Qure | *Frontend* | *Backend* **Node.js, Express.js, React Native, PostgreSQL, Socket.IO, Redis, Docker**

- Developed a full-stack clinic queue management system using **Node.js**, **TypeScript**, and a **Socket.IO** room architecture to push live queue updates to patients and staff, securing user workflows with role-based **JWT** authentication.
- Engineered fault-tolerant token management using atomic **PostgreSQL** transactions, and implemented a **BullMQ/Redis** caching pipeline that reduced asset upload latency by **70%**, containerizing the application via **Docker** for Render deployment.

PulseWatch | *Source Code* **React, Node.js, Server-Sent Events (SSE), GenAI, Twilio, Docker**

- Built a real-time system monitoring platform utilizing a **React** frontend and **Node.js** backend to track API and website health, streaming live updates via **Server-Sent Events (SSE)** and using **GenAI** to generate incident summaries for faster debugging.
- Implemented a multi-channel alerting pipeline routing production environment triggers to webhooks **(Slack, Discord)** and SMS (**Twilio**), significantly reducing team incident response times.

Skin Lesion Classification Platform | *App Repo* | *ML Model Repo* **Python, Flask, React (Vite), REST APIs**

- Trained a multi-task deep learning model on a highly imbalanced 7-class dataset, successfully achieving a **0.753 melanoma recall** and **0.928 AUC-ROC** to significantly outperform the 67% naive baseline.
- Deployed an end-to-end medical classification application by building a **Flask** backend and REST APIs to serve real-time model predictions to a responsive **React** frontend.

Certifications

AWS Certified Cloud Practitioner – Amazon Web Services | Verify May 2026

Leadership & Activities

Instructional Assistant (System Programming), Wilfrid Laurier University Jan 2026 – Apr 2026
Served as a technical problem solver for **50+ students** on Shell scripting and C programming, evaluating labs for code quality, memory management, and algorithmic readability.

Startup Build Hackathon 2026 by Sheryians Coding School | *2nd Runner-Up* Jun 2026
Built PulseWatch backend architecture in an agile, fast-paced hackathon environment under a strict 48-hour deadline.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C++, Kotlin, Bash/Zsh
Web & Backend: Node.js, Express, Spring Boot, React, Next.js, Vue.js, Svelte, Flask, Socket.IO, TanStack Query
Cloud & Infrastructure: AWS (EC2, S3), GCP, Docker, PostgreSQL, MongoDB, Redis, Git
Data & ML: PySpark, Pandas, NumPy, Scikit-Learn, Matplotlib